Karl Arvindraj

Software Engineer

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Summary

Full Stack Software Developer with hands-on experience building responsive web applications and RESTful APIs using Java, Spring Boot, React.js, TypeScript, and SQL. Experienced with Spring Security, JPA, JWT, PostgreSQL, MySQL, Supabase, and Git, backed by strong foundations in Data Structures and Algorithms, OOP, and DBMS. Focused on developing clean, scalable, maintainable, reliable, production-ready software.

Skills

Programming Languages: Java, JavaScript (ES6+), C, SQL
Frontend Technologies: HTML5, CSS3, React.js, TypeScript, Tailwind CSS
Backend Technologies: Spring Boot, Spring Security, RESTful APIs, JPA, JWT, Supabase
Databases & Core Concepts: PostgreSQL, MySQL, Data Structures and Algorithms, OOP, DBMS
Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, Postman, Vercel, Cloudflare Pages

Experience

Full-Stack Development Intern **September 2026 – Present**

ApexFlow Technologies *Tools Used: React.js, TypeScript, Supabase, PostgreSQL, Git, GitHub*

- Developing responsive client applications using React.js and TypeScript, building reusable components and interactive interfaces for live client requirements.
- Integrating Supabase and PostgreSQL for application data, authentication, and backend services across production client features.
- Collaborating with developers to implement requirements, debug issues, and deliver features across web and mobile environments.

Freelance Web Developer **February 2026 – Present**

Freelance *Tools Used: React.js, Git, GitHub, Vercel*

- Designed and developed responsive client websites using **React.js and JavaScript**, creating reusable components, interactive interfaces, and mobile-friendly layouts.
- Implemented advanced animations and interactive experiences using **GSAP, ScrollTrigger, and Lenis**, enhancing navigation, visual transitions, and overall user experience.
- Integrated functional contact forms and deployed production websites through **Vercel**, managing source code, updates, and development workflows with Git and GitHub.

Projects

BookMyDoc – Doctor Appointment Booking System **GitHub**

- Developed a **full-stack doctor appointment booking application** using **React, TypeScript, Java, and Spring Boot**, integrating frontend workflows with REST APIs for doctor discovery, appointment availability, and booking.
- Implemented **JWT-based authentication** using Spring Security and BCrypt password hashing to secure authenticated patient and doctor operations through stateless REST APIs.
- Designed relational database models for Users, Doctor Profiles, Slots, and Bookings using **Spring Data JPA, Hibernate, and MySQL**, establishing entity relationships and database constraints.
- Implemented appointment scheduling logic with backend validation and **overlapping-slot detection**, rejecting conflicting slot creation requests and preventing multiple bookings for the same slot through database constraints.

NovaPort AI – Trade Compliance Platform **GitHub**

- Engineered the user interface of a trade intelligence platform using **React and Tailwind CSS**, creating a seamless experience for document uploads and compliance checks.
- Integrated **Google Gemini AI** to automatically read and extract data from complex logistics documents, effectively replacing manual data entry processes.
- Integrated REST APIs with the **React frontend** to process AI-generated document data, enabling reliable data retrieval and dynamic compliance results.
- Implemented asynchronous state management to handle AI processing times, utilizing loading screens and error boundaries to ensure a smooth, crash-free user experience.

Education

B.Tech in Computer Science and Engineering **2023 – 2027**

SRM Ramapuram, Chennai *CGPA: 8.99 / 10*